

Check Your Progress

7. Define GIS.
8. What is remote sensing?
9. Name the two types of remote sensing.
10. What is an earthquake warning system?
11. Define flood warning.

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3.5 ROLE OF VARIOUS AGENCIES IN DISASTER MITIGATION: NATIONAL LEVEL AND STATE LEVELS

The general role of states (or governments) can be enumerated as follows:

- To provide relief to the people for the loss incurred due to disaster and to expedite rehabilitation and reconstruction and to ensure that all sections of the population are covered by efforts of the government
- The government also needs to put in place some social and economic activities for quick restoration of the situation
- To make efforts to minimize the impact of disasters by adopting precautionary measures and mechanisms
- To analyze and study the reasons for the occurrence of disasters and to suggest the remedies to avoid or minimize the effects of such natural calamities
- To make the best use of the funds, grants, donations and assistance received from the central government and other foreign countries or from other institutions/individuals for prevention of such natural calamities or for reconstruction and rehabilitation; to obtain loans and make proper use of the obtained funds

Institutional arrangements

In countries with federal system of government, various roles are distributed among the central and state governments. State governments further delegate their responsibilities down to the district authorities, block level as well as the village level. In fact, the Panchayat Raj Institutions in India are vested with specific roles in many administrative and developmental activities. However, the subject of disaster management has not been specifically mentioned in any of the three lists in the 7th schedule of the Indian Constitution, where subjects under the central and state governments that come under both are specified. On the legal front, there is no enactment either of the central or of any state government to deal with the management of disasters of various types in a comprehensive manner.

As a response to the numerous natural as well as man-made disasters, the governments of almost all the world nations have put institutional arrangements in place. India, too, has an integrated administrative machinery for management of disasters at the national, state, district and sub-district levels. The basic responsibility of undertaking rescue, relief and rehabilitation measures in the event of natural disasters, as at present, is that of the state governments concerned. The central government supplements the efforts of the states by providing financial and logistic support.

Central level

The level of response at the central government level is determined as per the current policy of relief expenditure fund and looking at factors like:

- Seriousness of the disaster
- Required scale of the relief and rescue operation
- Required central assistance as support for state-funded financial help and logistic support

The Contingency Action Plan (CAP) lists the initiatives that need to be taken by different central ministries and public departments after a natural disaster has taken place. The Plan will contain the procedures and identify the focal points in the administration to enable immediate launch of relief and rescue operations.

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The ministry of home affairs is the nodal ministry for coordination of relief and response and overall natural disaster management, and the department of agriculture and cooperation is the nodal ministry for drought management. Other ministries are assigned the responsibility of providing emergency support in case of disasters that fall in their purview.

The following decision-making and standing bodies are responsible for disaster management at the central level:

- Union Cabinet, headed by the Prime Minister
- Empowered Group of Ministers, headed by the Deputy Prime Minister
- National Crisis Management Committee (NCMC), under the chairmanship of the Cabinet Secretary
- Crisis Management Group (CMG): Under the chairmanship of the Central Relief Commissioner comprising senior officers from the various Ministries and other concerned departments which reviews contingency plans, measures required for dealing with a natural disaster, and co-ordinates the activities of the Central Ministries and the state governments in relation to disaster preparedness response and relief
- Technical organizations such as the Indian Meteorological Department (cyclone/earthquake), Central Water Commission (floods), Building Material and Technology Promotion Council (construction laws), Bureau of Indian Standards (norms), Defence Research & Development Organization (nuclear/biological), Directorate General Civil Defence provide specific technical support to coordination of disaster response and management functions
- The setting up of a National Disaster Management Authority (NDMA) is being contemplated by the Ministry of Home Affairs as the proposed apex structure within the government for the purpose. Amongst other major organizational initiatives, it is proposed to:
 - o Establish a specialised and earmarked response team for dealing with nuclear/biological/chemical disasters
 - o Establish search and rescue teams in each state
 - o Strengthen communication systems in the North-eastern region

Various agencies play various roles in disaster management-governments, NGOs, local agencies and the corporate sector. Some of these roles are overlapping. In addition, these agencies need to work in close cooperation with each other. In large countries like India, the government bodies function at many levels and through many different agencies. Therefore, coordination of their efforts is another challenge that needs to be met satisfactorily. Here, you will study about the roles of various agencies in disaster management.

State level

The responsibility to cope with natural disasters is essentially that of the state government. The role of the central government is supportive in terms of supplementation of physical and financial resources. The chief secretary of the state heads a state level committee which is in overall charge of the relief operations in the state and the relief commissioners who are in charge of the relief and rehabilitation measures in the wake of natural disasters in their states function under the overall direction and control of the state level committee. In many states, secretary, department of revenue, is also in-charge of relief. State governments usually have relief manuals and the districts have their contingency plan that is updated from time to time.

Table 3.2 Ministries Responsible for Disasters

Disasters	Nodal Ministry
Natural Disasters Management (Other than Drought Relief)	Ministry of Home Affairs
Drought Relief	Ministry of Agriculture
Air Accidents	Ministry of Civil Aviation
Railway Accidents	Ministry of Railways
Chemical Disasters	Ministry of Environment & Forests
Biological Disasters	Ministry of Health
Nuclear Disasters	Department of Atomic Energy

3.5.1 Role of Local Agencies

District and local level: As already discussed, the state governments are responsible for carrying out disaster management activities. However, the state governments get these activities carried out at the local level through district and local authorities. The district administration is the focal point for implementation of all governmental plans and activities. The actual day-to-day function of administering relief is the responsibility of the collector/district magistrate/deputy commissioner who exercises coordinating and supervising powers over all departments at the district level. Though it may not be a common phenomenon, there exists by and large in districts also a district level relief committee consisting of officials and non-officials.

The 73rd and 74th constitutional amendment have accorded the Panchayati Raj Institutions the status of 'Institutions of self-government'. The amendment has also laid down necessary guidelines for the structure of their composition, powers, functions, devolution of finances, regular holding of elections and reservation of seats for weaker sections including women. These local bodies can be effective instruments in tackling disasters through early warning system, relief distribution, providing shelter to the victims, and medical assistance.

Other local level agencies

Other than the national, state, district and local levels, there are various institutional stakeholders who are involved in disaster management at various levels in the country. These include the police and para-military forces, civil defence and home-guards, fire services, ex-servicemen, non-government organisations (NGOs), public and private sector enterprises, media and HAM operators (amateur radio operators), all of whom have important roles to play.

Armed forces

In India, the armed forces are called upon to intervene and take on specific tasks only when the situation goes beyond the control of civil administration. In practice, the armed forces are the core of the government's response capacity and tend to be the first responders of the central government in a major disaster. Due to their ability to organize action in adverse ground circumstances, speed of operational response and the resources and capabilities at their disposal, the armed forces have historically played a major role in emergency support functions such as communications, search and rescue operations, health and medical facilities, transportation, power, food and civil supplies, public works and engineering, especially in the immediate aftermath of disaster. Disaster management plans should incorporate the role expected of them so that the procedure for deploying them is smooth and quick.

International bodies

In the face of disasters, help usually pours in from various institutional and non-institutional agencies form within the nation as well as from outside. Foreign governments, especially the neighbours and other friendly nations are the first ones to offer aid. Many international bodies are vested with the responsibility to mitigate challenges posed by disasters. The Government of India is a member of various international organisations in the field of disaster response and relief. While, as a policy, no requests for assistance or appeals are made to the international community in the event of a disaster, assistance offered suo moto is accepted. Linkages exist with the following organizations:

- UN Office for Coordination of Humanitarian Affairs (UN OCHA), which has been made responsible by UN General Assembly mandate for all international disaster response
- United Nations Development Programme (UNDP), responsible for mitigation and prevention aspects of disaster management
- UN Disaster Assessment and Coordination (UNDAC) System

Community level initiatives

The goal of any disaster management initiative is to build a disaster resistant/resilient community equipped with safer living and sustainable livelihoods to serve its own development purposes. The community is also the first responder in any disaster situation, thereby emphasising the need for community level initiatives in managing disasters. To encourage such initiatives, the following are required:

- Creating awareness through disaster education and training and information dissemination are necessary steps for empowering the community to cope with disasters.

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- Community based approach followed by most NGOs and Community Based Organisations (CBOs) should be incorporated in the disaster management system as an effective vehicle of community participation.
- Within a vulnerable community, there exist groups that are more vulnerable like women and children, aged and infirm and physically challenged people who need special care and attention especially during disaster situations. Efforts are required for identifying such vulnerable groups and providing special assistance in terms of evacuation, relief, aid and medical attention to them in disaster situations.

Management of disasters should therefore be an interface between a community effort to mitigate and prevent disasters as also an effort from the government machinery to buttress and support popular initiatives.

Role of NGOs

The past few decades have witnessed the growth (and proliferation) of numerous NGOs in various fields. Although many NGOs devote their attention to specific issues, most of these swing into action in the face of a disaster. There are NGOs devoted to disaster management, which conduct research, carry out rehabilitation and reconstruction work when immediate help is not required.

NGOs are autonomous non-governmental bodies operating mostly as non-profit organizations. In India, they are very large in number and widely scattered across the country. Therefore, a need was felt to coordinate and streamline the activities of the NGOs so as to have the optimal benefit from their efforts. For this purpose, the National NGO Task Force has been set up. The main objective of this task force is to promote Community Based Disaster Risk Management (CBDRM) in all spheres of disaster management.

Some of the guiding principles of the task force are as follows:

- The National NGO Task Force on disaster management will be guided by CBDRM.
- This will also be the national-level platform for coordinating the activities of both government organizations and non-governmental organizations (GO-NGOs) in the field of disaster management.
- The Task Force will facilitate and support the formation of similar task forces at state and district levels.

Objectives of the task force

The objectives of the task force are as follows:

- Mapping of NGOs at state and district levels
- Training and capacity building of various stakeholders
- Mainstreaming vulnerability reduction through CBDRM in civil society initiatives
- Assisting the transition from relief codes to Disaster Management codes through Policy and Guidelines
- Guidelines for GO-NGO collaboration
- Providing assistance in setting up NGO task forces at state and district levels
- Integrating CBDRM in development programmes
- Strengthening disaster preparedness, mitigation and effective response in state and districts
- Setting up working groups on specific themes to include broad-based participation of civil society organizations

Advisory/advocacy role

- Coordination with government and other stakeholders in the field of disaster management
- Information sharing, documentation and dissemination, knowledge sharing, networking and technical support
- Training and capacity building
- Helping INGOs/NGOs articulate their CBDRM policies

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The task force has laid down several criteria for membership. Some of these criteria are as follows:

- A maximum of 20 members in the National Task Force
- The secretariat for the National NGO Task Force will be hosted at NDMA
- International, national, local NGOs and networks which are active at national, state and district level, undertaking disaster management initiatives in preparedness, capacity building, mitigation and response, including reconstruction programmes with potential for replication
- Organizations committed to CBDRM and having scope for wider dissemination and replication of CBDRM practices
- Organizations which have responded to a few major disasters
- Having the ability to dialogue and liaise with government and other stakeholders on disaster management and related policy issues
- Having demonstrated ability in training and capacity building

The list of institutions to be the members of the Task Force

- UNDP
- CARE, India
- OXFAM
- EFICOR
- SEEDS
- PGVS
- ABCD
- TNTRC
- RVC
- RedR India
- CARITAS
- World Vision
- Indian Red Cross Society
- BJS
- UNICEF
- MHA, GoI
- NIDM
- Ramakrishna Mission
- VANI
- SPHERE India

One of the most effective agencies during a disaster are the NGOs operating in the concerned area. These NGOs along with other local bodies (government assistance is desirable) can create community level initiative since community level initiatives are most effective.

3.5.2 Role of Corporate Agencies

Disasters are unforeseen events that cause great damage, destruction and human suffering. They require immediate, coordinated and effective response from multiple sources such as government, voluntary and corporate sector organizations. The corporate sector in India has played a valuable role in relief, recovery, rehabilitation and reconstruction.

An enhanced, expanded, well-defined and pro-active role is expected from corporates in all aspects of the disaster management cycle.

The NDMA recognizes the Indian corporate sector as one of the key stakeholders and envisages the involvement of this sector in awareness generation, disaster preparedness and mitigation planning through sensitization, training and co-opting of the corporate sector and their nodal bodies in planning process and response mechanism.

The corporate sector has the potential of assisting both the business community in protecting itself and the community at large in increasing its resilience to disasters.

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Objectives for corporate sector's involvement in disaster management

- To identify the roles of various corporate entities in different areas of disaster management, including disaster preparedness, mitigation, emergency response (relief, rehabilitation, reconstruction) and recovery
- To explore the potential roles of trade associations for mobilizing the corporate sector in disaster management
- To explore the full potential of corporate social responsibility and public private partnership in disaster management in India for professionalizing various aspects of governance
- To set up a national corporate task force on disaster management
- The corporates in India are constantly committed to extending all necessary help toward disaster management. Various corporate bodies coordinate their efforts. Meetings and follow-up meetings are called with other corporate associations such as FICCI, CII, ASSOCHAM and NASSCOM to discuss the modalities on public- private participation in disaster management, to explore the potential roles of trade associations in disaster management initiatives and to make strategies for setting up of a corporate task force for disaster management.

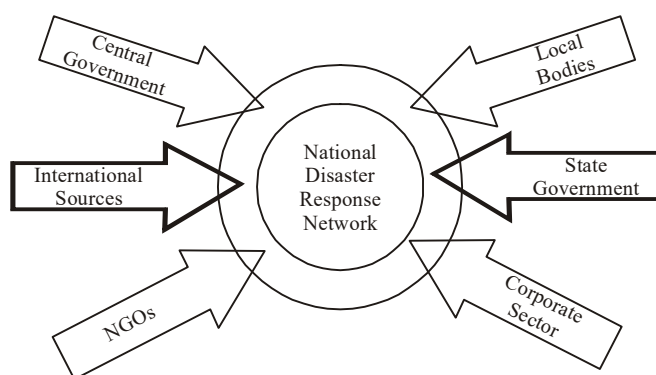


Fig. 3.2 National Disaster Response Network

3.5.3 Role of the Armed Forces

India is a very vulnerable nation which is prone to multiple natural disasters owing to its unique topographic and climatic conditions. Considering its size and dimensions, it has various different climates and geographical types. The coastal states, especially the ones on the eastern coast and Gujarat are exposed to cyclones where 40 million hectares (eight per cent) of land mass is flood prone, 68 per cent faces drought threat, 55 per cent of the area is in seismic zones III and IV and falls under earthquakes-prone belt and sub-Himalayan region and Western Ghats are threatened by landslides. India is increasingly getting affected by man-made disasters related to industrialization, transportation, environmental degradation and terrorist attacks. There is no legal ratification either at the central or the state governments level to deal with such disasters as it is not specified under any of the three lists (central, state and concurrent) of Seventh Schedule of the Indian Constitution.

The government of India is well aware that there is an urgent need for better disaster response mechanism, but the overall trend has indicated that the level of preparation to handle such disasters by the centre as well as the states is very unbalanced and unorganized which requires considerable strengthening. The centre and a number of states have started gearing up for such disasters which are becoming common by the day. The nodal agency for coordination of relief, response and overall natural disaster management has been positioned under the Union ministry of home affairs to design strategies to overcome such disasters and their effects. Whenever any disaster occurs in India, the armed forces are requested to intervene and take control of the situation.

Since the Indian civil administration is not adequately equipped for undertaking such urgent disaster response activities, they rely on the support given by the armed forces. On the other hand, Indian armed forces are one of the most technologically developed, dedicated and modernized armed forces in the world, adequately equipped with the necessary technical competence, man power and material resources undertakes rescue and relief operations of any disasters. An instance can be stated

when the tsunami that occurred in December 2004 where the Indian Army, Navy and the Air Force coordinated by the Integrated Defence Staff (IDS) efficiently executed the relief, rescue and evacuation assignments under Operation Sea Wave, and also extended aid to Sri Lanka and Maldives under Operation Rainbow and Operation Castor at the request of their respective governments for assistance. A few other such instances where the defence showed its presence was the Kashmir earthquake of 2005, the cyclone in Bangladesh on 15 November 2007, the fire breakout at Burrabazar in Kolkata on 12 January 2008 or the recent serial blasts at Bangalore and Ahmedabad in July 2008, the role played by the Indian armed forces is commendable.

The primary role being to defend the nation against any external forces the Indian armed forces are inevitably involved in securing the country from diverse unconventional fronts. They are always ready to execute any kind of operation related to any kind of disaster-affected areas and the daring and skill required to be able to work under adverse ground and climatic conditions. This continual performance in quick rescue and response action after disasters has been outstanding and with the ever increasing frequency of disasters in the South-Asian region, they continue to play a major role in the coming years.

In spite of the disaster rescue and relief responsibilities, there is need for decisive *modus operandi* for operational coordination between the civil administration and the armed forces. The armed forces' instant response depends on the information made available by the state administration, where any delay in information would cause loss of precious time to relieve the situation. In regard to this quick response the state should appreciate the immediate organizing and course of action in which the armed forces function. The armed forces' professional ethics are autonomous in character and do not encourage civil interference. As disaster management plan should incorporate the role of the armed forces so that the procedure for deploying them is well-organized, and there is an interface personnel correlation between the state government and the armed forces for immediate effective delivery of relief to the affected victim.

The armed forces has trained resources locations with rehearsals to:

- Carry out search and rescue operations to save the injured and the disposal of the deceased victims to prevent spread of epidemics
- Mapping of vulnerable areas and escape routes
- Building mechanisms of early warning, educating the public in safety measures are added components to the police functioning

Armed forces rescue hundreds in flood-hit states

New Delhi, Oct 5: Undertaking one of the biggest relief and rescue operations in the flood-ravaged states of Karnataka and Andhra Pradesh, the armed forces have rescued hundreds of marooned people so far.

"1,336 people have been rescued so far by the armed forces in the flood-ravaged states of Karnataka and Andhra Pradesh. The army, air force and the navy are working round-the-clock to rescue the marooned people and provide much needed food in the flood affected areas," a defence ministry spokesperson said Monday.

In the flood fury that hit Andhra Pradesh, Karnataka and other neighbouring states of Goa and Maharashtra Oct 1, following torrential rains under the influence of a low pressure area in the Bay of Bengal, a large number of towns and villages were either submerged or inundated by water from the Krishna, Tungabhadra and Don rivers.

"The army has deployed around 1000 of its men in Karnataka and Andhra Pradesh along with life jackets, boats and medical teams," the spokesperson added.

The flood situation is improving in Mahbubnagar district of Andhra Pradesh.

"The Indian Air Force (IAF) in one of its biggest rescue operations so far has deployed 29 aircraft and helicopters, including seven Chetaks and 10 MI-8s, for rescue and relief

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operations. These aircraft have already carried out 255 sorties and dropped a huge quantity of food packets in the affected areas,” an IAF official said.

The Indian Navy also has deployed 15 teams of its divers and experts to rescue the marooned people in the two states. Of these, five teams are at Vijaywada, three at Mahbubnagar, three at Kurnool in Andhra Pradesh, two at Karwar and two at Gadag district in Karnataka. These teams have already rescued more than 500 people from the affected areas.

Source: Reported by IANS : <http://www.newkerala.com/nkfullnews-1-125154.html>

When such natural calamities such as floods and drought occur they are addressed by the already existing number of plan schemes under which a lot is planned and can be done. The state governments need to implement and make full use of the existing plan schemes and prioritize the implementation of such schemes which would help in overcoming the hard hit situation left behind by the disaster. In certain cases when the damage is colossal, they even imply possible diversion of the funds from other schemes to those schemes which are implemented to ease such situations and provide immediate relief. In such crisis situations there may also be need for certain re-appropriations/reallocations among the different departments.

When the Indian armed forces are called upon to control the situation, they come with the basic requirements of the people in such situations. In practice, the armed forces are the main government's response capacity and are generally the first rescue team in situations of such severe crisis. Because of their ability to organize action in adverse ground circumstances with speed and efficiency, the Armed Forces play a major role in emergency support functions such as communications, search and rescue operations, health and medical facilities, transportation, power, food and civil supplies, public works and engineering, specially in the immediate aftermath of disaster. Disaster management plans should incorporate the role expected of them so that the procedure for deploying them is smooth and quick as displayed by the armed forces.

Although we have very efficient and swift acting armed forces for our rescue, there are some areas of concern that need to be worked upon, streamlined and require improvement urgently. They are:

- Integrated planning for disasters which also includes the intervention and roles of the armed forces formations into disaster management planning at all levels starting from the district level right up to the Central Government.
- Setting up of a permanent modernized well equipped national command centre or operations room, with advanced communications and data links to all state capitals. The national command centre or operations room needs to be operated on a 24-hour basis by professionals for instant response and resource integration. There needs to be a properly equipped operations room at the state level as well.
- A quick action team composed of experienced professionals, from both military and civilian backgrounds drawn from Central and state government staff to respond immediately and be present at the locations when a disaster strikes. This team needs to be organized and run professionally on the same basis as the United Nations Disaster Assessment and Coordination (UNDAC) teams.
- A set of urban search and rescue capacity at all levels, by establishing a fully equipped Search and Rescue team as a part of every fire department unit in all state capitals, with sufficiently trained staff and modern equipment such as thermal imagers, acoustic detection devices. This is of immediate relevance since a major weakness exposed in the Gujarat earthquake was a lack of specialized urban search and rescue capability in India.
- The media policy should be well geared to handle the growing real time television reporting which generates enormous political pressures on the government to respond rapidly and efficiently. This needs attention as it displays the immediate action taken in case of such diverse situations.
- An interface with a better understanding of the international system for disaster response and putting the required systems in place for dealing with international assistance such as customs, immigration, foreign policy implications. A closer weave is required between the Ministry of External Affairs and the relevant inter-national agencies concerned with disaster response.

- The standard procedures for dealing with domestic humanitarian and relief assistance from non-government sources need to be formulated. The procedures need to be laid out to avoid confusion and ensure the best resource utilization being offered, just as it happens in the case of international assistance.
- Modern unified legislation for disaster management. Keeping in mind the current division of responsibilities between the state and Central government, central and concurrent lists, there is a need to create a body of legislation dealing with response to natural disasters and other emergencies, clearly delineating responsibilities and powers of each entity and specifying what powers or actions would need to be triggered on declaration of a disaster by the Government of India or a state government. This legislation also needs to incorporate the current legislation dealing with emergencies caused by chemicals that need to be dealt by the Ministry of Environment so that all emergencies are dealt with under one law. The legislation should include clear definitions of what constitutes a disaster at a national level.

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3.5.4 Role of Government Agencies

Each department and government agency involved in disaster management and mitigation will do the following:

- Designate a nodal officer for emergency response who will act as the contact person for that department/agency
- Ensure establishment of fail-safe two-way communication with the state, district and other emergency control rooms and within the organization
- Emphasis on communication systems used regularly during LO with more focus on the use of VHF's with automatic repeaters, mobile phones with publicized numbers, HF radio sets. It should be remembered that SAT phones fail during prolonged emergencies and electric failure if the phones cannot be recharged.
- Work under the overall supervision of the SRC/the district Collectors during emergencies.

Let us look at the way in which various agencies manage disasters:

1. Agriculture

Prevention activities

- Awareness generation regarding various plant diseases, alternate cropping practices in disaster-prone areas, crop insurance, provision of credit facilities, proper storage of seeds
- Hazard area mapping (identification of areas endemic to pest infections, drought, flood, and other hazards)
- Develop database village-wise, crop-wise, irrigation source wise, insurance details, credit facilities
- Regular monitoring at block level; the distribution and variation in rainfall. Prepare the farmers and department officers to adopt contingency measures and take up appropriate course of action corresponding to the different emerging conditions
- Detail response manuals to be drawn up for advising the farmers for different types of disasters, e.g., rain failure in July or September and development of a dynamic response plan taking into account weekly rainfall patterns
- Develop IEC materials to advise the farming communities on cropping practices and precautionary measures to be undertaken during various disasters
- Improving irrigation facilities, watershed management, soil conservation and other soil, water and fertility management measures keeping in mind the local agro-climatic conditions and the proneness of the area to specific hazards
- Promotion of alternative crop species and cropping patterns keeping in mind the vulnerability of areas to specific hazards
- Surveillance for pests and crop diseases and encourage early reporting
- Encourage promotion of agro service outlets/enterprise for common facilities, seed and agro input store and crop insurance

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Preparedness activities

- Review and update precautionary measures and procedures and especially ascertain that adequate stock of seeds and other agro inputs are available in areas prone to natural calamities
- Review the proper functioning of rain gauge stations, have stock for immediate replacement of broken/non-functioning gadgets/equipments, record on a daily basis rainfall data, evaluate the variation from the average rainfall and match it with the rainfall needs of existing crops to ensure early prediction of droughts

Response activities

- Management of control activities following crop damage, pest infestation and crop disease to minimize losses
- Collection, laboratory testing and analysis of viruses to ensure their control and eradication
- Pre-positioning of seeds and other agro inputs in strategic points so that stocks are readily available to replace damage caused by natural calamities
- Rapid assessment of the extent of damage to soil, crop, plantation, irrigation systems, drainage, embankment, other water bodies and storage facilities and the requirements to salvage, re-plant, or to compensate and report the same for ensuring early supply of seeds and other agro inputs necessary for re-initiating agricultural activities where crops have been damaged
- Establishment of public information centres with appropriate and modern means of communication, to assist farmers in providing information regarding insurance, compensation, repair of agro equipments and restarting of agricultural activities at the earliest

Recovery activities

- Arrange for early payment of compensation and crop insurance dues
- Facilitate provision of seeds and other agro inputs
- Promotion of drought and flood-tolerant seed varieties
- Review with the community, the identified vulnerabilities and risks for crops, specific species, areas, which are vulnerable to repetitive floods, droughts, other natural hazards, water logging, increase in salinity, pest attacks and draw up alternative cropping plans to minimize impacts to various risks
- Facilitate sanctioning of soft loans for farm implements
- Establishment of a larger network of soil and water testing laboratories
- Establishment of pests and disease monitoring system
- Training in alternative cropping techniques, mixed cropping and other agricultural practices which will minimize crop losses during future disasters

2. Water Supplies and Sanitation (Public Health Engineering and Rural Water Supply and Sanitation)

Prevention activities

- Provision of safe water to all habitats
- Clearance of drains and sewerage systems, particularly in the urban areas

Preparedness activities

- Prior arrangement of water tankers and other means of distribution and storage of water
- Prior arrangement of stand-by generators
- Adequate prior arrangements to provide water and halogen tablets at identified sites to be used as relief camps or in areas with high probability to be affected by natural calamities
- Raising of tube-well platforms, improvement in sanitation structures and other infrastructural measures to ensure least damages during future disasters
- Riser pipes to be given to villagers

Response activities

- Disinfections and continuous monitoring of water bodies
- Ensuring provision of water to hospitals and other vital installations
- Provision to acquire tankers and establish other temporary means of distributing water on an emergency basis
- Arrangement and distribution of emergency tool kits for equipments required for dismantling and assembling tube wells
- Carrying out emergency repairs of damaged water supply systems
- Disinfection of hand pumps to be done by the communities through prior awareness activities and supply of inputs

Recovery activities

- Strengthening of infrastructure
- Review and documentation
- Sharing of experiences and lessons learnt
- Training to staff
- Development of checklists and contingency plans

3. Police

Prevention activities

- Keep the force in general and the ODRAF in particular fighting fit for search, rescue, evacuation and other emergency operations at all times through regular drills
- Procurement and deployment of modern emergency equipments while modernizing existing infrastructure and equipments for disaster response along with regular training and drills for effective handling of these equipments
- Focus on better training and equipments for ODRAF for all types of disasters, e.g. diving equipments
- Rotation of members of ODRAF so that the force remains fighting fit
- Ensure that all communication equipments including wireless are regularly functioning and deployment of extra wireless units in vulnerable pockets
- Ensure interchangeability of VHF communication sets of police and OSDMA supplied units, if required.
- Keeping close contact with the District Administration and Emergency Officer
- Superintendent of Police be made Vice Chairperson of District Natural Calamity Committee
- Involvement of the local army units in response planning activities and during the preparation of the annual contingency plans to ensure logistics and other support to armed forces during emergencies

Response plan

- Security arrangements for relief materials in transit and in camps
- Senior police officers to be deployed in control rooms at State and district levels during L 1 level deployment onwards
- Deploy personnel to guard vulnerable embankments and at other risk points
- Arrangement for the safety
- Coordinate search, rescue and evacuation operations in coordination with the administration
- Emergency traffic management
- Maintenance of law and order in the affected areas
- Assist administration in taking necessary action against hoarders, black marketers

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4. Civil Defence

Prevention activities

- Organize training programmes on first-aid, search, rescue and evacuation
- Preparation and implementation of first aid, search and rescue service plans for major public events in the state
- Remain fit and prepared through regular drills and exercises at all times

Response activities

- Act as support agency for provision of first aid, search and rescue services to other emergency service agencies and the public
- Act as support agency for movement of relief
- Triage of casualties and provision of first aid and treatment
- Work in co-ordination with medical assistance team
- Help the police for traffic management and law and order

5. Fire services

Prevention activities

- Development/enforcement of relevant legislations and regulations to enhance adoption of fire safety measures
- Modernization of fire-fighting equipments and strengthening infrastructure
- Identification of pockets, industry which are highly susceptible to fire accidents or areas, events which might lead to fires, building collapse and educate people to adopt safety measures. Conduct training and drills to ensure higher level of prevention and preparedness
- Building awareness in use of various fire protection and preventive systems
- Training the communities to handle fire emergencies more effectively
- VHF network for fire services linked with revenue and police networks
- Training of masons and engineers in fire proof techniques
- Making clearance of building plans by fire services mandatory

Response activities

- Rescue of persons trapped in burning, collapsed or damaged buildings, damaged vehicles, including motor vehicles, trains and aircrafts, industries, boilers and pressure vessels, trenches and tunnels
- Control of fires and minimizing damages due to explosions
- Control of other dangerous or hazardous situations such as oil, gas and hazardous materials spill
- Protection of property and the environment from fire damage
- Support to other agencies in the response to emergencies
- Investigation into the causes of fire and assist in damage assessment

6. Civil supplies

Preventive activities

- Construction and maintenance of storage godowns at strategic locations
- Stockpiling of food and essential commodities in anticipation of disaster
- Take appropriate preservative methods to ensure that food and other relief stock are not damaged during storage, especially precautions against moisture, rodents and fungus infestation

Response activities

- Management of procurement
- Management of material movement
- Inventory management

Recovery activities

- Conversion of stored, unutilized relief stocks automatically into other schemes like Food for Work. Wherever, it is not done leading to damage of stock, it should be viewed seriously

7. Works/Rural development departments

Prevention activities

- Keep a list of earth moving and clearing vehicles/equipments (available with government departments, PSUs, and private contractors) and formulate a plan to mobilize those at the earliest
- Inspection and emergency repair of roads/bridges, public utilities and buildings

Response activities

- Clearing of roads and establishing connectivity. Restore roads, bridges and where necessary make alternate arrangements to open the roads to traffic at the earliest
- Mobilization of community assistance for clearing blocked roads
- Facilitate movement of heavy vehicles carrying equipments and materials
- Identification and notification of alternative routes to strategic locations
- Filling of ditches, disposal of debris, and cutting of uprooted trees along the road
- Arrangement of emergency tool kit for every section at the divisional levels for activities like clearance (power saws), debris clearance (fork lifter) and other tools for repair and maintenance of all disaster response equipments

Recovery activities

- Strengthening and restoration of infrastructure with an objective to eliminate the factor(s) which caused the damage
- Review and documentation
- Sharing of experiences and lessons learnt
- Training to staff
- Development of checklists and contingency plans

8. Energy

Prevention activities

- Identification of materials/tool kits required for emergency response
- Ensure and educate on the minimum safety standards to be adopted for electrical installation and equipments and organize training of electricians accordingly
- Develop and administer regulations to ensure safety of electrical accessories and electrical installations
- Train and have a contingency plan to ensure early electricity supply to essential services during emergencies and restoration of electric supply at an early date
- Develop and administer code of practice for power line clearance to avoid electrocution due to broken/fallen wires
- Strengthen high-tension cable towers to withstand high wind speed, flooding and earthquake, modernize electric installation, strengthen electric distribution system to ensure minimum damages during natural calamities
- Conduct public/industry awareness campaigns to prevent electric accidents during normal times and during and after a natural disaster

Response activities

- Disconnect electricity after receipt of warning
- Attend sites of electrical accidents and assist in undertaking damage assessment
- Stand-by arrangements to ensure temporary electricity supply

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- Prior planning and necessary arrangements for tapping private power plants like those belonging to ICCL, NALCO, RSP during emergencies to ensure uninterrupted power supply to the Secretariat, SRC, OSDMA, Police Headquarters, All India Radio, Doordarshan, hospitals, medical colleges, Collectorate Control Rooms and other vital emergency response agencies
- Inspection and repair of high tension lines /substations/transformers/poles
- Ensure the public and other agencies are safeguarded from any hazards, which may have occurred because of damage to electricity distribution systems
- Restore electricity to the affected area as quickly as possible
- Replace/restore of damaged poles/salvaging of conductors and insulators

9. Water resources department

Prevention activities

- Assess preparedness level
- Annual assessment of danger levels and wide publicity of those levels
- Identify flood prone rivers and areas and activate flood monitoring mechanisms
- Provide water level gauge at critical points along the rivers, dams and tanks
- Identify and maintain of materials/tool kits required for emergency response
- Stock-pile of sand bags and other necessary items for breach closure at the Panchayat level

Response activities

- Monitoring flood situation
- Dissemination of flood warning
- Ensure accurate dissemination of warning messages naming GPs and Tehsils with details of flow and likely damage
- Monitoring and protection of irrigation infrastructures
- Inspection of bunds of dams, irrigation channels, bridges, culverts, control gates and overflow channels
- Inspection and repair of pumps, generator, motor equipments, station buildings
- Community mobilization in breach closure

Recovery activities

- Strengthening of infrastructure and human resources
- Review and documentation
- Sharing of experiences and lessons learnt
- Training of staff
- Development of checklists and contingency plans

10. Fisheries

Prevention activities

- Registration of boats and fishermen
- Building community awareness on weather phenomena and warning system especially on Do's and Don'ts on receipt of weather related warnings
- Assist in providing life saving items like life jackets, hand radios
- Certifying the usability of all boats and notifying their carrying capacities
- Capacity building of traditional fishermen and improvisation of traditional boats which can be used during emergencies
- Train up young fishermen in search & rescue operation and hire their services during emergency

Response activities

- Ensure warning dissemination to fishing communities living in vulnerable pockets

- Responsible for mobilizing boats during emergencies and for payment of wages to boatmen hired during emergencies
- Support in mobilization and additional deployment of boats during emergencies
- Assess the losses of fisheries and aquaculture assets and the needs of persons and communities affected by emergency

Recovery activities

- Provide compensations and advice to affected individuals, community
- Plan for rehabilitation in the long run of the fisherfolk including resettlement, insurance, better boats, nets, etc., improved communication system as a part of the communication hub for warning dissemination, especially for marine fishermen and fishing communities residing in high-risk areas

11. Forest department

Prevention activities

- Promotion of shelter belt plantation
- Publishing for public knowledge details of forest cover, use of land under the forest department, the rate of depletion and its causes
- Keep saws (both power and manual) in working conditions
- Provision of seedling to the community and encouraging plantation activities, promoting nurseries for providing seedlings in case of destruction of trees during natural disasters

12. Transport department

Prevention activities

- Listing of vehicles which can be used for emergency operation
- Safety accreditation, enforcement and compliance
- Ensuring vehicles follow accepted safety standards
- Build awareness on road safety and traffic rules through awareness campaign, use of different IEC strategies and training to school children
- Ensure proper enforcement of safety regulations response activities
- Requisition vehicles, trucks, and other means of transport to help in the emergency operations
- Participate in post impact assessment of emergency situation
- Support in search, rescue and first aid
- Failure to cooperate and misappropriation of relief materials to invite disqualification from the post

Recovery activities

- Provision of personal support services e.g. counselling
- Repair/restoration of infrastructure e.g. roads, bridges, public amenities
- Supporting the GPs in development of storage and in playing a key role and in the coordination of management and distribution of relief and rehabilitation materials the Panchayat Samity and GP members to be trained to act as an effective interface between the community, NGOs, and other developmental organizations
- Provide training so that the elected representatives can act as effective supportive agencies for reconstruction and recovery activities

13. Panchayati raj

Preventive activities

- Develop prevention/mitigation strategies for risk reduction at community level
- Training of elected representatives on various aspects of disaster management
- President, Zilla Parishad to be made member of District Natural Calamity Committee

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- Public awareness on various aspects of disaster management
- Organize mock drills
- Promote and support community-based disaster management plans
- Support strengthening response mechanisms at the GP level (e.g., better communication, local storage, search & rescue equipments)
- Clean drainage channels, organize through community participation trimming of branches before cyclone season
- Ensure alternative routes/means of communication for movement of relief materials and personnel to marooned areas or areas likely to be marooned
- Assist all the government departments to plan and prioritize prevention and preparedness activities while ensuring active community participation

Response activities

- Train the GP members and support for timely and appropriate delivery of warning to the community
- Clearance of blocked drains and roads, including tree removal in the villages
- Construct alternative temporary roads to restore communication to the villages
- PRLs to be a part of the damage survey and relief distribution teams to ensure popular participation
- Operationalize emergency relief centres and emergency shelter
- Sanitation, drinking water and medical aid arrangements
- IEC activities for greater awareness regarding the role of trees and forests for protection during emergencies and also to minimize environmental impact which results owing to deforestation like climate change, soil erosion
- Increasing involvement of the community and NGOs in plantation, protection and other forest protection, rejuvenation and restoration activities
- Plan for reducing the incidence, and minimize the impact of forest fire
- Assist in road clearance
- Provision of tree cutting equipment
- Provision of building materials such as bamboos etc. for construction of shelters

Recovery activities

- Take up plantation to make good the damage caused to tree cover
- Information and Public Relations Department

Prevention activities

- Creation of public awareness regarding various types of disasters through media campaigns
- Dissemination of information to public and others concerned regarding do's and don'ts of various disasters.
- Regular liaisoning with the media

Response activities

- Setting up of a control room to provide authentic information to public regarding impending emergencies
- Daily press briefings at fixed times at state and district levels to provide official version (during LO also)
- Media report and feedback to field officials on a daily basis from L1 onwards
- Keep the public informed about the latest of the emergency situation (area affected, lives lost)
- Keep the public informed about various post-disaster assistances and recovery programmes

3.5.5 Role of Media in Disaster Management

Disasters can be both natural and man-made. But the root causes of some of the seemingly natural disasters may also be certain human activities carried on in utter disregard of their consequences to the nature. Such natural disasters are therefore preventable. Since all man-made disasters and some of the so-called natural disasters are preventable, the media can educate and forewarn the people about the consequences of their dangerous actions and operations. More and in-depth education on the subject becomes necessary where the human activities and the natural calamities they lead to, are separated by a period of time. In such cases, though the causal connection is direct, since the consequences occur at a distant point of time, people fail to appreciate the link between the two and continue to indulge in their depredations on nature, digging in turn sometimes slowly, sometimes fast, a grave for humanity.

The floods, droughts and water famine situations are many times directly traceable to the human activity, while drainage mismanagement and air and water-pollution, environmental destruction and global warming are all clearly on account of the man's misdeeds. Some excavations and destruction of forests are responsible for landslides and mudflows, while according to experts some earthquakes are caused by the construction of the large dams and by impounding large quantities of water in them. The dam failures, dam bursts, mine fires, epidemics, food poisoning, chemical and industrial disasters, nuclear disasters and all accident-related disasters are undoubtedly the handiwork of man.

The impending occurrence of some natural disasters whether induced by human actions or otherwise, can now be known sufficiently in advance, thanks to the advances in science and technology. The media, by communicating the information to the people and the concerned authorities sufficiently in advance, can enable them to take the necessary steps to prevent and minimize the losses of lives and property.

While the disaster is on, the media can also play the role of relaying the measures that are being taken and monitoring them, cautioning the affected or to be affected people about the Dos and Don'ts, of scotching rumours and preventing panic and confusion, of establishing contacts, of identifying the needy spots and focusing attention on them, and generally by assisting the authorities, voluntary organizations and volunteers in reaching, informing and assuring the affected ones of the assistance and the measures taken, for their relief. During the onslaught of the disaster, what is of utmost importance is to keep the morale of the people high, to create self-confidence in them, to prevent panic and to maintain order by assuring and making available the necessary help readily and quickly. The media can help in many ways in ensuring these conditions.

The rescue, relief and rehabilitation measures need an integrated and co-coordinated approach and for that purpose all agencies, government and non-government, have to pool their resources together for efficient, expedient and effective work on all fronts. The collection of material resources and the enlisting of man-power are as much important as their efficient utilization. The depiction of devastation and of human misery through the media many times by itself acts as an appeal to the people to come forward to render help in various ways. In addition, the specific appeal made for relief through the columns and the time-slots of the media, brings in sizeable aid in the requisite form. At the same time, it becomes necessary to keep a watch and report on some anti-social elements who try to take advantage of such situations.

Some disasters like floods, cyclones and droughts have become a periodic feature of our national life. The vulnerable spots and the sections of the populace also stand identified over the years. Almost always, the worst sufferers are the weaker sections of the society. They are unable to shift from these places, because there lie their sources of livelihood and all that they have in life to preserve and protect. They constitute a vast section of our society, and in normal conditions they contribute in sizeable proportion to our national wealth. Yet, except in the times of disaster, they are rarely remembered and the measures for the permanent solution of their plight are hardly ever discussed in the media. The media can also focus its attention on this problem.

Not insignificant is the contribution the media can make in countering the damaging, exaggerated and negative reporting and propaganda in the foreign media on the occurrence of the disasters. This country has witnessed such phenomena in the past. A prompt presentation of the real state of affairs by our media including the news-agencies, and the correction of the misrepresentations by them will go a long way in dispelling the wrong impressions created abroad which may otherwise have adverse effects on the administration, the economy and the polity of the country.

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On some occasions in the past the media fulfilled its duties in full measure while on other occasions it was found wanting. However, if the media has to discharge its role properly on such occasions, it is also necessary to have a complete cooperation between it and all governmental and non-governmental agencies. The extent of the co-ordination and co-operation between them determines the nature, the degree and the scale of the preparation to prevent and meet the disasters.

People across the globe are more disaster-conscious today than they ever were, thanks to the media. Countries that have a large geographical extent, house a large population, have a vast coastal strip and many mountainous areas and rivers, are more vulnerable to disasters. Prevention is better than cure and prevention can only happen through education and information dissemination. The role of the media in educating the population and providing information is of immense importance.

It is not just now that the media has come into prominence. Even in the past, Mahatma Gandhi made use of the media, that is, newspapers, to convey the methods of preventing plague. He wrote in the *Amrit Bazar Patrika*, about the techniques of preventing plague with the aim of educating the masses. Not only did he appeal to the people to adopt clean habits and ensure hygiene, he requested the educated groups to spread the word about the importance of hygiene and sanitation. His writings, popularized by the media, encouraged people to collect and dispose off garbage and use the pictures of Gods and Goddesses to prevent people from throwing garbage in public places.

When a calamity strikes, it is the responsibility of the media to motivate people to act collectively. A well-established print media in combination with television and radio can play a significant role in generating mass action during disasters. The media today has at its disposal all the tools and instruments for educating the public and enabling them to discharge their duties as well-informed and responsible citizens.

Media can help in many ways when disaster strikes. It can:

- Make people aware of mitigation efforts
- Guide them to utilize funds properly
- Act as an interface between the government and the people
- Help reduce panic
- Direct people on where essential services are available
- Provide counselling
- Help prevent corruption in the distribution of relief material
- Facilitate two-way communication between the provider of assistance and the affected

In times of a disaster or in any emergency, for that matter, it is very important for the government to maintain good relations with the media. It is important to give reporters information that is timely and accurate. It is also important to allow them access to emergency zones up to a certain limit. This helps them realize that honest efforts are being made to tackle emergency and prevents them from becoming adversarial.

The media can be informed through press releases or press conferences. Nowadays, information is also posted on websites. A press release should be used only when the urgency is not immediate. Otherwise, in fast-moving situations, where the site is swarming with reports and the crowd is getting difficult to handle, it makes more sense to arrange a press conference on the spot.

The information that is conveyed to the population through the media should have the following characteristics:

- It should be clear in stating what the people should expect in the situation and what they are expected to do.
- It should take into account the effect the news would have on the people.
- It should be based on the predicted response of the population.
- It should be able to provide guidance to the people.
- It should be based on valuable inputs from media experts.
- It should have the public good in mind.

There is a need for journalists to be more proactive in reporting on drought or disaster risk-related issues.

*Disaster Relief
Operations*

- Journalists need to be given regular training on preparing for disasters and for improving their understanding of the scope of the disaster, its hazards and the relevant key terms.
- The journalists should work closely with the policy makers so that positive stories are broadcast or printed. This, in turn, ensures that some kind of change is brought about and appropriate solutions can be offered to the disaster-prone areas.
- Journalists should be held accountable for what they report and the reactions their reports trigger.
- The advocacy stories that are broadcast or printed should target the ministries so that speedy action takes place.
- Workshops should be arranged for journalists so that they become proactive while reporting from the sites of disaster.
- There should be an exclusive team of editors working online to post the reports/stories, sent by journalists in the field, on to the website.
- Journalists should maintain a database of disaster-prone areas so that the impacts of hazards can be monitored.
- News agencies and magazines should announce contests and awards for disaster stories that are well-researched. This will ensure that the reports are genuine.

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Dimensions of communication situations

The media can support or hinder the disaster management activities of the government and relief agencies. Before trying to understand the importance of the media in controlling and mitigating disaster, one must first take a look at the three aspects of communication situations as follows:

- Audience
- Stage
- Level of effect
 - o It is very important to understand the audience what the media is addressing. During a disaster, there are clearly two distinct parts of the population. One part which is actually affected by the disaster and the other which is not.
 - o What is communicated should also take into account the stage of the situation. Before the disaster occurs, the media can communicate to the public, the ways in which disasters should be averted and managed. This information should help prepare the population. During a disaster, the media should give warning message; educate people on how best to handle the disaster if it occurs; and also provide reassurance. After the disaster, once again, the media should be able to provide the sources of help, information and medical aid. It should be able to counsel the affected and help them get over the trauma. At this stage, the media should also analyse the situation, help investigate what went wrong and ensure that some lesson is learnt from the disaster.
 - o The degree to which an individual or the population collectively is affected should be assessed by the media. They should also be able to tell how an individual can contribute to the relief work and how people collectively can not only help manage the situation but also educate others.

Hurdles in working with the media

Although we all know how important a role media can play in managing disasters, it is a fact that the media is not always easy to approach.

- The media may not find a situation as worthy of attention as it may be made out to be by the sources.
- The message may not be put across by the media in the right tone. There is always a chance for distortion.
- The media may not always succeed in painting a trustworthy or credible picture.

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- They may try to interfere where their presence is least required or even unwanted/harmful.
- Instead of being sympathetic, they may choose to be critical and distant.
- At times, they may communicate contradictory risk estimates.
- They may give more importance to sensationalism instead of concentrating on giving serious information.
- Most journalists work with the aim of keeping a critical eye on those wielding power.

Reducing the losses of life and property caused by natural hazards is a compelling objective now receiving worldwide attention. It is now being increasingly believed that the knowledge and technology base potentially applicable to the mitigation of natural hazards has grown so dramatically that it would be possible, through a concerted cooperative international effort, to save many lives and reduce human suffering, dislocation, and economic losses simply by better information, communication and awareness. Timely mass media communication about impending disasters can lead to appropriate individual and community action, which is the key to implementing effective prevention strategies including evacuation and survival of people. Such communications can educate, warn, inform, and empower people to take practical steps to protect themselves from natural hazards.

The role of media, both print and electronic, in informing the people and the authorities during emergencies thus, becomes critical, especially the ways in which media can play a vital role in public awareness and preparedness through educating the public about disasters; warning of hazards; gathering and transmitting information about affected areas; alerting government officials; helping relief organizations and the public towards specific needs; and even in facilitating discussions about disaster preparedness and response. During any emergency, people seek up-to-date, reliable and detailed information.

The main principle of information provision, therefore, should be an ethical one and so, during an emergency, the media should be sensitive to the needs of the public in affected areas and should avoid misinforming and broadcasting unconfirmed reports that may lead to despair and panic. Therefore, correct and reliable information disseminated through the media is an important instrument for balancing the possible effects of incorrect, misleading or even willfully distorted information. Reliable and timely information provided through the media can help people overcome any kind of fear and fatalism during and after an emergency. Indeed, the availability of reliable and timely information and knowledge about an event and the resulting needs help to improve solidarity and also creates an atmosphere conducive to collective response for sharing the humanitarian challenges created by disasters. Media today has arguably penetrated every household in the world, in one form or another. Journalists pride themselves in reporting objectively on global events and regard an independent media as one of the pillars of democratic society.

Mass media have certain characteristics that make them powerful instrument of disaster communication as they provide quick and easy access to large number of people located at different places; they highlight the problems and difficulties faced by the people affected by the disasters; they mobilize public opinion for humanitarian assistance. On the other hand, sources dealing with the media know that media can also prove to be difficult channels of communication. There is no direct control over the content and form of information transmitted and sometimes there are competing and even conflicting information which may always be very congenial for mobilizing support for concerted action. Contrary to popular perceptions journalists may not always be independent but may act as 'gate keepers, interpreters and commentators' representing larger business interests of the media owners. A closer examination of the global media illustrates that it is not one homogeneous entity pursuing a uniform method in disseminating their views. On the contrary, there are a small number of large factions which compete with one another to increase their respective audiences and commercial gains. Media ethics, therefore, become an important aspect of media reporting during any disaster event. It assists media workers in determining what is right and choosing the best from several alternatives. Ethics should set guidelines, rules, norms, codes and principles to lead journalists and other media workers to make moral decisions.

Check Your Progress

12. State any three objectives of the National NGO Task force.
13. List any three recovery activities of Water Supplies and Sanitation.
14. Mention any three recovery activities of the transport department.
15. State the ways in which media can help when disaster strikes.
16. What are the three aspects of communication situations?

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3.6 SUMMARY

- One of the main objectives of development programs in many developing countries of the world for the past several decades is poverty reduction.
- The aim of Community- based disaster management (CBDM) is to reduce vulnerabilities and strengthen people's capacity to cope with hazards.
- The Government of India has adopted mitigation and prevention as essential components of their development strategy. The Tenth Five Year Plan document has a detailed chapter on disaster management.
- A GIS is defined as a system that is designed to gather, store, manipulate, analyse, manage and present all types of geographical data.
- GIS captures, interprets and transforms geographical data into graphical outcomes using advanced hardware and software technology.
- Acquiring information about a remote object or phenomenon, without making physical contact with the object is called remote sensing.
- Remote sensing is a very useful tool of GIS.
- Spatial technology is an integrated hardware and software system which brings together GIS and remote sensing to use satellites for a number of uses. Spatial technology plays an important role in disaster management.
- An earthquake warning system is a system of accelerometers, communication, computers, and alarms that is devised for regional notification of a substantial earthquake while it is in progress.
- The responsibility to cope with natural disasters is essentially that of the state government. The role of the central government is supportive in terms of supplementation of physical and financial resources.
- The goal of any disaster management initiative is to build a disaster resistant/resilient community equipped with safer living and sustainable livelihoods to serve its own development purposes.
- NGOs are autonomous non-governmental bodies operating mostly as non-profit organizations.
- Three stages of communication situations as follows— audience, stage, and level of effect.
- The role of media, both print and electronic, in informing the people and the authorities during emergencies thus, becomes critical, especially the ways in which media can play a vital role in public awareness and preparedness through educating the public about disasters; warning of hazards; gathering and transmitting information about affected areas; alerting government officials, helping relief organizations and the public towards specific needs; and even in facilitating discussions about disaster preparedness and response.

3.7 KEY TERMS

- **Community involvement:** Community involvement refers to a 'less than' ideal situation, where the community is asked to participate in a program that has already been designed by someone else.